PawPal: A Pet Care Mobile App

Project Team

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Chapter 1

Introduction

PawPal is a comprehensive mobile application designed to revolutionize the pet care ecosystem by integrating health, community, and commerce into a single unified platform. Pet owners face challenges managing their pets' health records, vet appointments, dietary needs, social engagement, and marketplace transactions across fragmented applications. PawPal addresses these issues by creating an all-in-one pet care solution that delivers convenience, personalized care, and active community involvement.

1.1 Problem Statement

Pet owners struggle to effectively manage multiple aspects of pet care due to scattered apps addressing isolated needs—such as health tracking, veterinary booking, lost pet alerts, and pet product marketplaces. The fragmented nature of existing solutions limits pet safety, reduces efficiency in care, and complicates access to reliable local support. Additionally, lack of structured and secure systems for adoption and recovery of lost pets creates significant risks and anxiety among pet families.

1.2 Motivation

The motivation for PawPal lies in transforming the pet care experience by providing a single, user-friendly application that covers all needs. By integrating advanced AI-driven pet identification, personalized care plans, real-time community support, and secure commerce, PawPal empowers pet owners to focus on their pets' well-being without juggling multiple platforms. Enhanced local support through vet connectivity and community events further adds to safety and engagement. The motivation also extends to ethical and secure trading, promoting responsible pet ownership.

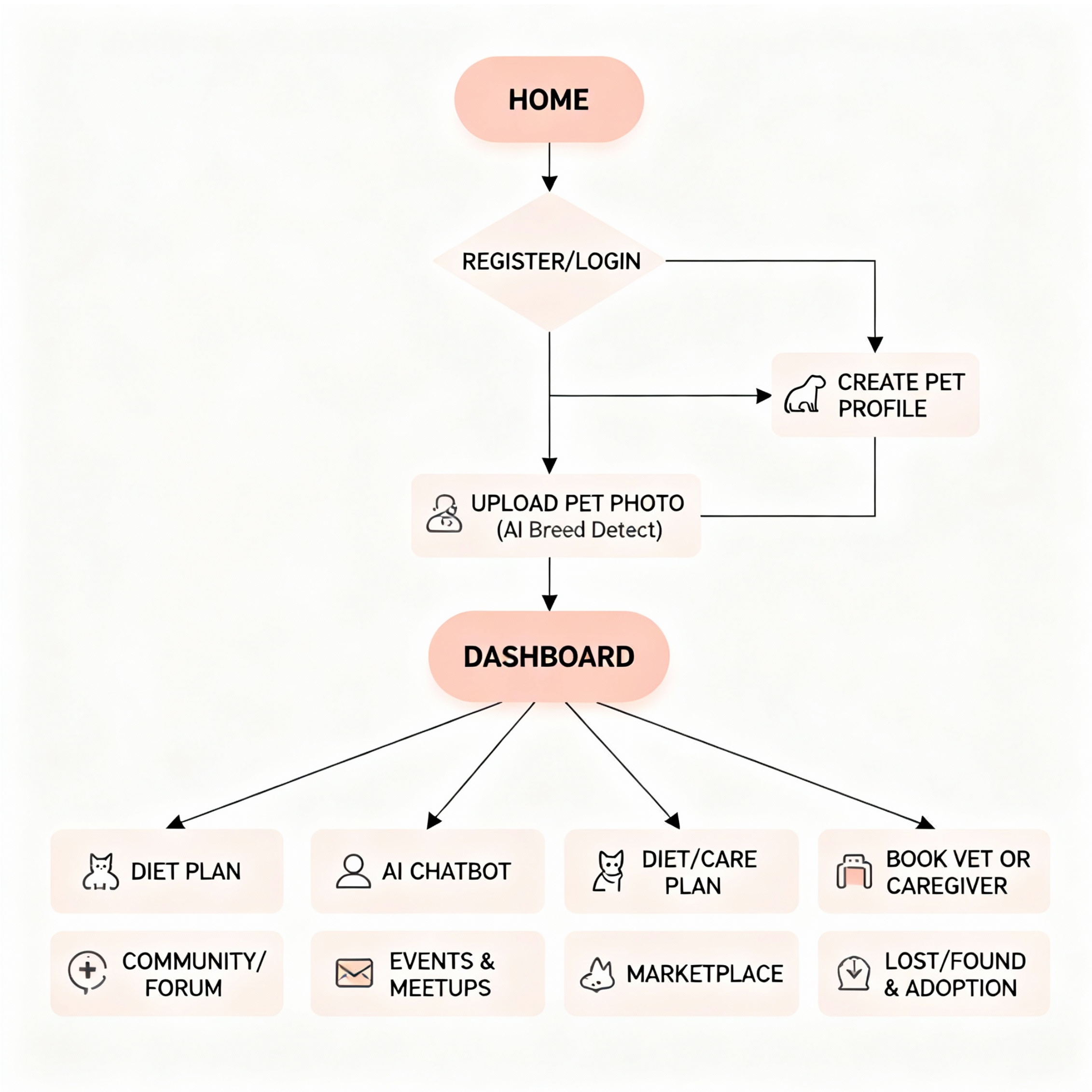

Figure 1.1: Project high level view

1.3 Problem Solution

PawPal proposes a smart, all-in-one platform combining features like breed detection through AI, AI-powered chatbot for vet guidance and diet plans, pet profiles and journals, secure pet toy marketplace, social community hubs for lost and found alerts, event planning, and caregiver services including GPS tracking. Integrated veterinary services and payment gateways (Stripe and crypto wallets) enable seamless health care and commerce transactions. The platform aims to deliver a global peer-to-peer network embedded with local support structures, making pet care simpler, safer, and more connected.

1.4 Stake Holders

- Pet Owners: Primary users managing pet health, social activities, and shopping.
- Veterinarians and Caregivers: Providers of clinical care, emergency support, and pet sitting/walking services.
- Pet Product Sellers: Marketplace participants offering toys and accessories.
- Community Members: Users involved in forums, events, lost & found networks.
- Supervisors and Developers: Overseeing and implementing the project.

Chapter 2

Project Description

2.1 Scope

The scope of PawPal encompasses developing a cross-platform mobile application aimed at unifying various pet care functions into a seamless user experience. It covers AI-driven pet breed identification, a 24/7 veterinary chatbot, personalized diet and care management, community engagement through forums and lost & found networks, a safe and secure pet toy marketplace, event planning for pet-friendly activities, services for pet caregivers with tracking capabilities, and integrated payment solutions supporting traditional and cryptocurrency channels. This mobile application will leverage Flutter for frontend development, robust backend APIs using Flask and Node.js, and utilize relational and NoSQL databases for storing user, pet, and transaction data. The app will cater to both local and global users, balancing social engagement with commercial functionality.

2.2 Modules

The system is divided into the following major modules:

2.2.1 Pet Identification Module

- AI-based breed detection from pet images.

- Provides breed-specific care and diet recommendations.

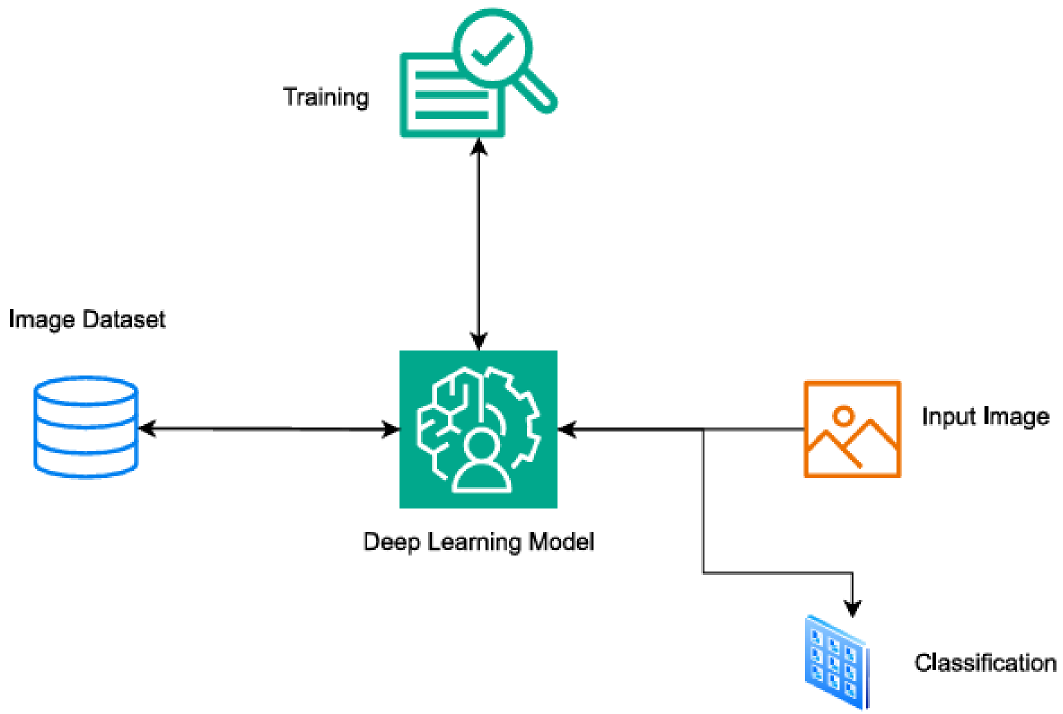

Figure 2.1: Breed image classifier

2.2.2 Chatbot Module

- AI-powered veterinary guidance and diet plan generation.
- Provides 24/7 support for health queries.

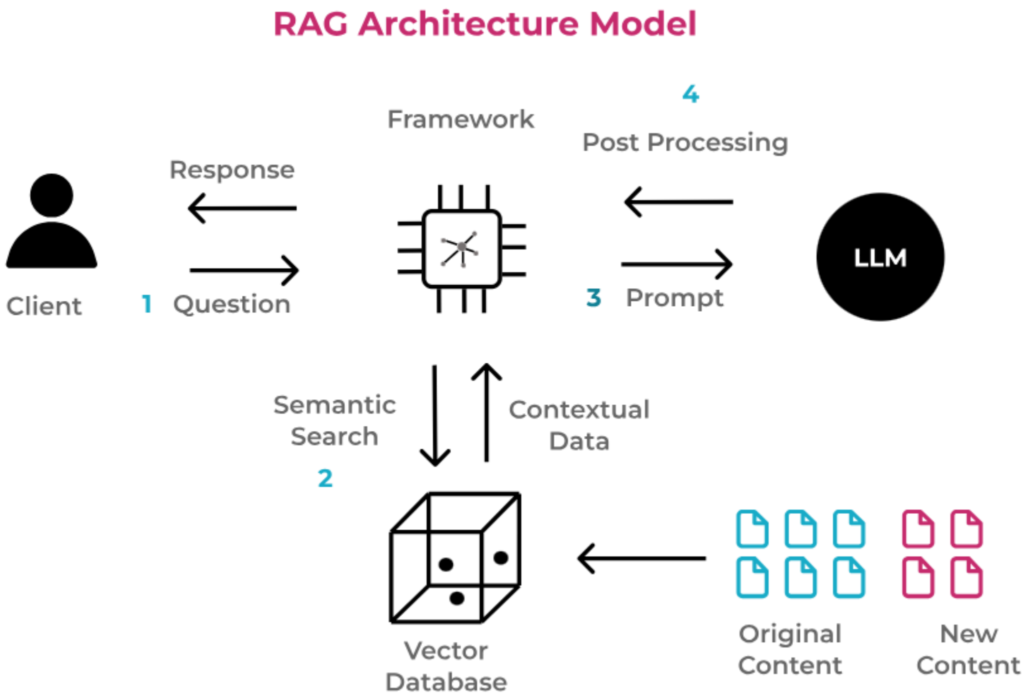

Figure 2.2: Chatbot

2.2.3 Pet Profiles & Journals

- Allows owners to create and maintain digital pet profiles.
- Includes health journals and activity logs.

2.2.4 Pet Toy Marketplace

- Secure platform for buying and selling pet toys.
- Supports seller/buyer profiles and transaction histories.

2.2.5 Community Hub

- Real-time forum posting, comments, media sharing.
- Lost and found pet network and adoption modules.
- Event and meetup planning features.

2.2.6 Veterinary Services Integration

- Instant vet appointment booking and consultation.
- Nearby vet location and 24/7 online vet support.

2.2.7 Pet Caregiver Services

- Profile verification for caregivers.
- Booking and GPS tracking of pet walks and sitters.

2.2.8 Payment Integration

Supports Stripe and cryptocurrency wallets for marketplace and services.

2.3 Tools and Technologies

- Frontend: Flutter for cross-platform mobile development ensuring responsive UI and smooth UX.
- Backend: Python Flask and Node.js REST APIs for business logic and data management.
- Database: PostgreSQL and MongoDB for structured user/pet data and document storage.
- AI/ML: CNN-based breed detection, RAG and LLM for chatbot intelligence.
- Payments: Crypto wallet APIs for secure financial transactions.
- Geolocation: OpenStreetMap for event and service location tracking.
- Notifications: Push notifications for care reminders, lost pet alerts, and events.

2.4 Work Division

Table 2.1: Work Division

Name	Registration	Responsibility / Module / Feature
Hamza Naveed	22i-0961	Lead UI/UX design and frontend development using Flutter and Figma; prototyping and responsive layouts; client-side validation; collaborate on notifications, vet booking, and marketplace checkout flows.
Huzaifa Nasir	22i-1053	Backend API design and development with Flask/Node.js; AI pet identification model training and integration; chatbot backend development and maintenance; authentication, authorization, real-time security middleware.
Maaz Ali	22i-1042	AI model training, testing, and integration; vet booking and caregiver services backend; social and commerce backend modules including community hub, lost & found, pet marketplace, payment gateway; admin panel features.

2.5 Timeline

For each module and respective feature, responsibility is assigned to a team member.

Table 2.2: Table 2: Project Timeline

Iteration #	Time Frame	Tasks / Modules
01	Sept–Oct	User authentication, pet profile creation, initial pet identification module, UI wireframes, backend API base structure.
02	Nov–Dec	Enhanced pet identification, launch chatbot, diet & care plan generator, veterinary booking system, API endpoints, basic chat UI.
03	Jan–Feb	Community hub launch (forums, media uploads), secure pet toy marketplace, lost & found and adoption modules, seller/buyer profiles, admin dashboard.
04	Mar–Apr	Event planner, caregiver services with profile verification and GPS tracking, final marketplace polish, admin moderation and reporting tools.

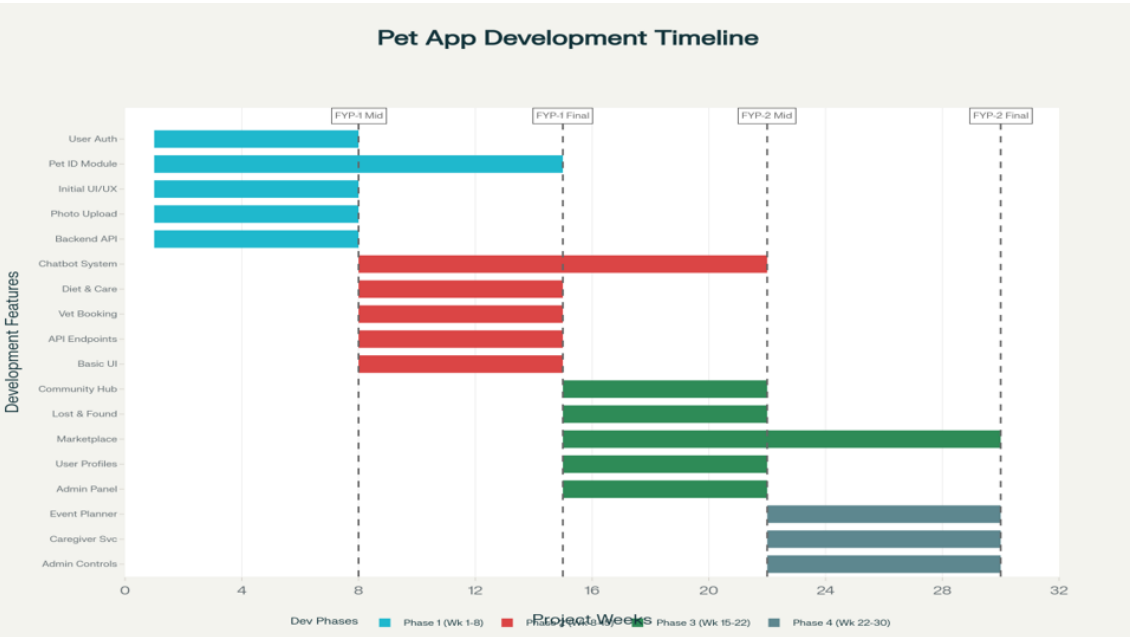

Figure 2.3: Project Gantt chart as per the timeline